

Homework 1: Multi-Qubit Gates, Entanglement, and Teleportation

Quantum Computing Bootcamp 2026

These problems are meant to be self-contained. A classical bit has value 0 or 1. A qubit is described by a normalized vector in a complex vector space. For example,

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

When a qubit is measured in the computational basis $\{|0\rangle, |1\rangle\}$, the probability of outcome 0 is $|\alpha|^2$ and the probability of outcome 1 is $|\beta|^2$.

Problem 1: Inner Products and Orthogonality

A one-qubit state can be represented as a column vector

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha |0\rangle + \beta |1\rangle, \quad |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The inner product of two states $|u\rangle = \begin{bmatrix} u_0 \\ u_1 \end{bmatrix}$ and $|v\rangle = \begin{bmatrix} v_0 \\ v_1 \end{bmatrix}$ is

$$\langle u|v\rangle = u_0^* v_0 + u_1^* v_1,$$

where $*$ means complex conjugation. Two states are orthogonal if their inner product is 0.

- (a) Compute $\langle 0|0\rangle$, $\langle 1|1\rangle$, $\langle 0|1\rangle$, and $\langle 1|0\rangle$. Use your answers to explain why $\{|0\rangle, |1\rangle\}$ is called an orthonormal basis.
- (b) Let

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Write $|+\rangle$ and $|-\rangle$ as column vectors. Then compute $\langle +|+\rangle$, $\langle -|-\rangle$, and $\langle +|-\rangle$. What do these calculations say about normalization and orthogonality?

- (c) Consider the states

$$|\psi\rangle = \frac{1}{2} |0\rangle + \frac{\sqrt{3}}{2} |1\rangle, \quad |\phi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{i}{\sqrt{2}} |1\rangle.$$

Verify that both states are normalized. Then compute $\langle \phi|\psi\rangle$.

Problem 2: XOR, CNOT, and the Toffoli gate

The XOR operation, written $\oplus$, outputs 1 exactly when its two input bits differ:

$$0 \oplus 0 = 0, \quad 0 \oplus 1 = 1, \quad 1 \oplus 0 = 1, \quad 1 \oplus 1 = 0.$$

The controlled-NOT gate, or CNOT, acts on two bits or qubits. The first input is called the control and the second input is called the target:

$$|c, t\rangle \mapsto |c, t \oplus c\rangle.$$

For basis states ordered as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, this rule gives

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

- (a) Apply the CNOT matrix to each computational basis vector $|00\rangle, |01\rangle, |10\rangle$, and $|11\rangle$. By linearity, gates act on superpositions one term at a time. Compute

$$\text{CNOT} \left(\frac{|00\rangle + |10\rangle}{\sqrt{2}} \right).$$

Is the resulting state a product state or an entangled state? Give a brief explanation.

- (b) Construct a two-qubit circuit made from one-qubit gates and CNOT gates that transforms

$$|00\rangle \mapsto \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle), \quad |11\rangle \mapsto \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

What does your circuit do to $|01\rangle$ and $|10\rangle$? Build your circuit in Qiskit. Test your circuit on the inputs $|00\rangle$ and $|11\rangle$ and print the resulting statevectors.

- (c) The Toffoli gate, also called the controlled-controlled-NOT gate, acts on three bits or qubits. The first two inputs are controls and the third input is the target:

$$|a, b, t\rangle \mapsto |a, b, t \oplus (a \cdot b)\rangle.$$

Make a truth table for all eight computational basis inputs. Identify exactly which inputs are changed by the Toffoli gate. Explain how the Toffoli gate can compute the classical AND of two bits a and b if the target starts in $|0\rangle$.

Problem 3: Tensor products of states and matrices

Multi-qubit states are built using tensor products. For one-qubit states

$$|a\rangle = a_0|0\rangle + a_1|1\rangle = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}, \quad |b\rangle = b_0|0\rangle + b_1|1\rangle = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix},$$

the tensor product is

$$|a\rangle \otimes |b\rangle = \begin{bmatrix} a_0b_0 \\ a_0b_1 \\ a_1b_0 \\ a_1b_1 \end{bmatrix} = a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle.$$

This problem uses the ordering $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

- (a) Compute the column vectors for $|00\rangle = |0\rangle \otimes |0\rangle$, $|01\rangle = |0\rangle \otimes |1\rangle$, $|10\rangle = |1\rangle \otimes |0\rangle$, and $|11\rangle = |1\rangle \otimes |1\rangle$.
- (b) Expand $|+\rangle \otimes |+\rangle$ in the computational basis. What is the probability of each two-bit measurement outcome?
- (c) Expand $|+\rangle \otimes |-\rangle$ in the computational basis. What changes compared with part (b)? Which probabilities change, and which amplitudes change?
- (d) For matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix},$$

the tensor product is

$$A \otimes B = \begin{bmatrix} aB & bB \\ cB & dB \end{bmatrix}.$$

Compute $X \otimes I$ and $I \otimes X$, where

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

- (e) Apply $X \otimes I$ and $I \otimes X$ to the basis vector $|01\rangle$. Explain the difference between flipping the first qubit and flipping the second qubit.

Problem 4: Quantum Teleportation

In lecture, we saw a brief conceptual overview of the teleportation protocol. In this problem, we will demonstrate, mathematically, how it works. Alice starts with an unknown qubit

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle.$$

Alice and Bob also share an entangled Bell pair

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Use the qubit order

Alice's unknown qubit Alice's Bell-pair qubit Bob's Bell-pair qubit.

Thus the initial three-qubit state is

$$|\psi\rangle \otimes |\Phi^+\rangle = (\alpha |0\rangle + \beta |1\rangle) \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

- (a) Expand the initial three-qubit state in the computational basis.
- (b) Alice applies CNOT with her unknown qubit as control and her Bell-pair qubit as target. Write the resulting state three-qubit state.

- (c) Alice applies a Hadamard gate to her unknown qubit. Use

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Write the resulting three-qubit state.

- (d) Group your answer from part (c) according to Alice's two measurement bits. Show that the state can be written as

$$\frac{1}{2} \left(|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\beta|0\rangle + \alpha|1\rangle) + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\beta|0\rangle - \alpha|1\rangle) \right),$$

where the first two qubits are Alice's and the final qubit is Bob's.

- (e) For each possible pair of Alice measurement outcomes 00, 01, 10, and 11, identify Bob's post-measurement state. Then identify which correction Bob should apply from the set $\{I, X, Z, XZ\}$ to recover $|\psi\rangle$.
- (f) Implement the teleportation circuit in Qiskit for the test state

$$|\psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}.$$

Simulate the circuit with statevectors or measurement counts. Verify that Bob's final qubit matches the input state after the appropriate corrections.

Bonus Problem: A guided CHSH inequality experiment

This bonus problem is longer than the others. The CHSH inequality is one of the cleanest ways to see that entanglement can produce correlations that are stronger than any classical local hidden-variable model allows. CHSH-type experiments are part of the family of Bell tests. The 2022 Nobel Prize in Physics was awarded to Alain Aspect, John Clauser, and Anton Zeilinger for experiments with entangled photons, establishing violations of Bell inequalities which demonstrated that quantum mechanics cannot be described by a (secret) deterministic classical theory.¹

In a CHSH experiment, Alice and Bob share many identical pairs of entangled qubits. For each pair:

- Alice chooses one of two measurement settings, called A_0 and A_1 .
- Bob chooses one of two measurement settings, called B_0 and B_1 .
- Each measurement returns one of two values: $+1$ or -1 .

Here, the word *setting* means the measurement choice used on a particular qubit before reading out the result. Physically, this could be a detector orientation or a measurement axis. In a circuit model, it often corresponds to applying a basis-change gate and then measuring in the computational basis. For example, the pair of settings (A_0, B_1) means: Alice measures her qubit using her first measurement choice, while Bob measures his qubit using his second measurement choice.

For each pair of settings, define the correlation

$$E(A_i, B_j) = P(\text{same outcomes}) - P(\text{different outcomes}).$$

¹See the official Nobel Prize press release: <https://www.nobelprize.org/prizes/physics/2022/press-release/>.

The CHSH combination is

$$S = E(A_0, B_0) + E(A_0, B_1) + E(A_1, B_0) - E(A_1, B_1).$$

A classical local hidden-variable model must satisfy

$$|S| \leq 2.$$

Quantum mechanics predicts that suitable measurements on a Bell state can give

$$|S| = 2\sqrt{2},$$

which is larger than 2.

- (a) In your own words, describe the experimental setup. What do Alice and Bob choose? What do they record? Why do they need many repeated trials rather than a single pair of qubits?
- (b) Suppose Alice and Bob use fixed settings A_i and B_j for many trials. Let

$$N_{++}, N_{+-}, N_{-+}, N_{--}$$

be the numbers of trials with the corresponding outcomes. Write a formula for estimating $E(A_i, B_j)$ from these four counts. Then compute E for the sample data

$$N_{++} = 420, \quad N_{+-} = 80, \quad N_{-+} = 70, \quad N_{--} = 430.$$

- (c) Here is the short classical argument behind the CHSH bound. Imagine that, before measurement, each pair secretly carries predetermined answers

$$a_0, a_1, b_0, b_1 \in \{+1, -1\}$$

for Alice's two possible settings and Bob's two possible settings. For one pair, define

$$s = a_0b_0 + a_0b_1 + a_1b_0 - a_1b_1.$$

Factor this as

$$s = a_0(b_0 + b_1) + a_1(b_0 - b_1).$$

Show that s must equal either $+2$ or -2 . Then explain why averaging over many pairs gives $|S| \leq 2$.

- (d) Now use the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

First consider measuring both qubits in the Z basis. What outcomes can appear? What does this say about the correlation between Alice and Bob for this one measurement basis?

- (e) For the Bell state $|\Phi^+\rangle$, measurements in the x - z plane have correlation

$$E(\hat{a}, \hat{b}) = a_x b_x + a_z b_z,$$

where $\hat{a} = (a_x, a_z)$ and $\hat{b} = (b_x, b_z)$ are unit vectors describing Alice's and Bob's measurement axes in the x - z plane. Use the four settings

$$A_0 = Z, \quad A_1 = X, \quad B_0 = \frac{Z + X}{\sqrt{2}}, \quad B_1 = \frac{Z - X}{\sqrt{2}}.$$

Compute

$$E(A_0, B_0), \quad E(A_0, B_1), \quad E(A_1, B_0), \quad E(A_1, B_1),$$

and then compute S .

- (f) Compare your value of S with the classical CHSH bound $|S| \leq 2$. Entangled pairs of qubits violate this classical bound – entanglement cannot be described by assuming that each pair of qubits secretly carries predetermined answers!
- (g) Suppose a quantum device claims to prepare entangled pairs for Alice and Bob. Describe a simple CHSH-based test for nonclassical correlations in the device. Your answer should say what Alice and Bob repeatedly do, what data they collect, how they compute S , and what conclusion they can draw if they estimate $|S| > 2$ by a statistically convincing margin.
- (h) Implement a simple CHSH simulation in Qiskit. Prepare $|\Phi^+\rangle$ using H on qubit 0 followed by `cx(0, 1)`. Estimate the four correlations from measurement counts and compute S . One way to measure the observable

$$\cos(\theta)Z + \sin(\theta)X$$

is to apply `ry(-theta)` before measuring in the computational basis. Use angles

$$A_0 : \theta = 0, \quad A_1 : \theta = \frac{\pi}{2}, \quad B_0 : \theta = \frac{\pi}{4}, \quad B_1 : \theta = -\frac{\pi}{4}.$$

Print your four estimated correlations and your estimate for S . Your answer should be close to $2\sqrt{2}$ when the number of shots is large.